How-To Library

Informatica

Informatica MDM SaaS Match and Merge Best Practices Guide

© Copyright Informatica LLC 2023. Informatica, the Informatica logo, and Informatica Cloud are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

This article contains information about the best practices to configure match and merge. These best practices have been derived from real-world experiences that cover different use cases.

Supported Versions

- Informatica MDM SaaS August 2023

Table of Contents

Overview. . . . 2
Prerequisites. . . . 3
Key terminologies 3
Before you begin. . . . 3
Candidate selection criteria. . . . 4
Factors to consider before configuring key generation and candidate search levels. . . . 5
Key generation level. . . . 5
Candidate search level. . . . 6
Example scenarios for candidate selection criteria configuration. . . . 7
Match population. . . . 8
Merge strategy. . . . 8
Merge threshold. . . . 9
Guidelines to configure declarative rules. . . . 9
Segment matching. . . . 10
Test and validate match rules. . . . 10
Tuning exercise. . . . 10
Survivorship rules. . . . 10
Challenges. . . . 11
Do's and don'ts of match and merge configuration. . . . 11
Training a match model. . . . 11
Recommendations for selecting training data sets. . . . 12
Recommendations for training machine learning models. . . . 12
Match and merge best practices checklist. . . . 12

Overview

MDM SaaS is a cloud-based master data management platform that you can use to manage records. You can configure the match and merge process to identify and resolve duplicate records. To identify duplicates, you can choose to use a machine learning (ML) model, declarative match rules, or both.

Prerequisites

Before the Informatica team can provision the MDM SaaS environments, ensure that your IT administrator completes the required prerequisite questionnaires and checklist.

Key terminologies

The following table lists the terms that Informatica Intelligent Data Management Cloud uses:

Term	Description
Source	Location from where you retrieve data.
Target	Location to which you move the data.
Connection	Provides access to data in cloud and on-premise applications, platforms, databases, and flat files.
Org	A secure area within the Informatica Intelligent Data Management Cloud repository that stores your licenses, user accounts, assets, such as business entities and reference data. You might have access to one or more organizations.
Secure Agent	A lightweight program that runs all tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services.
Asset	An object that can be created, read, updated, and deleted in Informatica Intelligent Data Management Cloud.

Before you begin

Before you configure the match and merge process, audit the characteristics of your data and the specific match and merge requirements.

Ensure that you have the following information about business data:

- Gain a comprehensive understanding of data sources, data quality, and the specific attributes that are required to match. Analyze the uniqueness and variability of data values to determine the attributes to include in match rules.
- Analyze the onboarded data and check if matching and merging duplicate data is required.
- Determine the size of the data set which could impact the performance of the match and merge process.
- Leverage data profiling techniques to gain insights into data patterns, anomalies, and data quality issues. This analysis helps identify potential match rule improvements and uncover hidden relationships within your data. Data profiling allows you to understand the data thus enabling you to make better decisions on the creation of fuzzy match keys and match rules.
- Identify the attributes that are most critical for matching records accurately. These attributes include names, addresses, phone numbers, email addresses, or any other relevant fields that hold unique identifiers or distinct patterns.

The following table describes different tasks that you can perform to audit data:

Task	Description
Get data samples	- Get data samples from different data sources. - Ensure that the data sample represents the production data, so avoid data from test systems. - The size of the data sample can be as big as you can analyze.
Talk to users	- Discuss with the users to determine the high-level match requirements or rules. - Understand what is a match and not a match for them based on the sample data set. - Check for any samples from previous systems that they can share.
Analyze data samples	- Discover fields that you can use to create match keys. - Use SQL queries, data profiler, or other tools to analyze data. - Assess the quality of each field and combination of fields. - Analyze the data samples and prepare clear guidelines with examples on what fields to use for matching and what not to use.
Identify groups	Run simple group statistics on single or multiple field data. The group statistics help you identify the following details: - Large group of identical records. - Levels of exact match duplicates. Based on it, you can determine whether to use exact or fuzzy rules. - Appropriate candidates selection criteria. A large number of candidates might impact match performance. - Determine viability of the intended match key fields.
Verify data completeness	Determine how often a field has a non-null value and look for data completeness by fields and combination of fields. For example, determine the total number of records that have both first and last names values. If the postal code value is present in only 50% of the records, it might not be a suitable candidate as an exact match field.
Determine data quality	- Ensure that the data is accurate. For example, gender field must contain only gender values. - Use data quality rule associations to refine the data. - Use pattern analysis to assess the quality of data in a field. Pattern analysis is useful in analyzing data that conforms to certain formats or data types. For example, check the format of the postal codes. - Standardize data. For example, use JR for junior. - Standardize and clean your address information.
Suspect data	Look for extraneous data that you can remove. To detect unnecessary data, you can browse through the data or use a data profiler tool.

Candidate selection criteria

Match candidates are record pairs that are possible matches. The criteria for selecting match candidates depends on business entity fields.

Before configuring candidate selection criteria, consider the following guidelines:

- If data contains organization names, use organization name to generate match keys. If data contains individual names, use person name to generate match keys. If data contains addresses, use address part 1 to generate match keys.
- Use multiple fields to generate match keys if they can identify more candidates that might not be identified if you use a single field.

- Use match key width that gives you the best performance without affecting your match quality. You might have to trade off some of the match quality for increased performance.
- Use the narrowest possible search level to generate acceptable matches. In most cases, the typical search level might suffice.
- Look for large groups of identical keys due to identical records. For example, high frequency phone numbers, email addresses, words, or phrases can return identical keys. You might need to clean or suppress data that returns large candidate lists.
- Consider filtering candidates to minimize the number of candidates for large data volumes. For example, you can filter candidates using state code or country code.

Factors to consider before configuring key generation and candidate search levels

Before you configure the key generation and candidate search levels, perform the following checks:

- How big is the target data set going to be?
- How important is performance over undermatching??
- Is the data quality good?
- Do you need to cleanse the data?
- Do you need to flag the suspect records?
- Do you need to add an additional candidate selection criteria?
- Do you need to change the key width or type?
- Do you need to add a candidate filter?

Key generation level

The key generation level defines the thoroughness with which the match model generates match keys to identify candidates for matching. The key generation levels are extended, limited, and standard.

Decide on a key generation level based on the following considerations:

- Size and quality of the data
- Reliability of the matched records
- Processing time for key generation

The following table describes the key generation levels:

Key Generation Level	Description
Extended	Performs a stringent analysis of your data to generate many match keys, which might result in a large number of match candidates. This level ensures reliable matches, but at the cost of long processing time to generate match keys. If your data volume is not too large and the data quality is poor, use the extended key generation level.
Limited	Performs a quick and lenient analysis of your data to generate few match keys, which might result in few match candidates. This level trades some match reliability for a faster search of match candidates compared to when you use the extended key generation level. If your data volume is extremely large with variations in word order, use the limited key generation level. The results of the limited key generation level are a subset of the standard level.
Standard	Generates an optimum number of match keys, which in turn results in an optimum number of match candidates. This level balances the reliability of matches and the processing time to generate match keys.

The following table lists the key generation levels required for different types of data:

Key Generation Level	Thoroughness of Key	Processing Time	When to Use
Extended	High	More	- Small data set - Poor data quality
Limited	Low	Less	- Large data set - Good data quality
Standard	Optimal	Optimal	- Average volume of data - Average data quality

Candidate search level

The match model uses the candidate search level to determine how stringently and thoroughly to search for match candidates. Later, the match model applies the declarative match rules on the match candidates and not on the entire data. You can configure an exhaustive, extreme, narrow, or typical candidate search level.

The match model uses the key generation level and the candidate search level that you define to determine the records that are potential match candidates.

Decide on a candidate search level based on the following considerations:

- Size and quality of the data
- Criticality of the matches
- Time constraints

The following table describes the candidate search levels:

Candidate Search Level	Description
Extreme	Performs a lenient search that returns many more match candidates than the exhaustive search level. Appropriate for small volumes of data that is not reliable and incomplete, or if it is critical to find the highest possible number of match candidates. You can also use the extreme level for the initial phase of matching and use a more stringent search level for the later phase. This search level results in a larger number of match candidates than the other search levels, possibly resulting in overmatching, and processes slowly.
Exhaustive	Performs a lenient search that returns many match candidates. Appropriate for small volumes of data that is not reliable and incomplete. You can also use the exhaustive level for the initial phase of matching and use a more stringent search level for the later phase. This search level can result in more match candidates than most other search levels, possibly resulting in overmatching, and processes slowly.
Typical	Use in most scenarios to search for an optimum number of match candidates.
Narrow	Performs a stringent search that returns fewest match candidates. Appropriate for large volumes of data that is reliable and complete or contains a large number of duplicates. This search level results in fewer match candidates than other search levels, possibly resulting in undermatching, and processes faster.

The following table lists the candidate search levels required for different types of data:

Candidate Search Level	Number of Candidates	Processing Time	When to Use
Extreme	Very high	The most	- Small data set - Highly critical data
Exhaustive	High	More	- Small data set - Critical data
Typical	Optimal	Optimal	- Average volume of data - Moderately critical data
Narrow	Few	Less	- Very large data set - Less critical data

Example scenarios for candidate selection criteria configuration

When you configure candidate selection criteria, consider the industry, the size and quality of data, the consequences of missing a match or overmatching, and the processing time available.

The following scenarios explain the usage of recommended key generation and search level settings for better matching:

- A flight reservation system needs to check if a passenger is on the no-fly list. It's a critical operation and lives might be at stake if a match is missed. In this scenario, an extended key generation results in the largest number of keys and an extreme search level returns as many candidates as possible. The process might take a while to run, but time is less important than missing a match.
- A retail customer says they are already registered for the loyalty program and the cashier is searching for their account. The cashier might expect a faster response because other customers might be waiting in line.

Missing a match might make this one customer unhappy, but that might not be too serious. In this scenario, using a standard key generation and a typical search level might result in quick matching. You can have a good balance of performance and results.

- A financial institution is merging customer records. Because of the sensitivity of this data, incorrectly merging two different customers is the worst possible outcome. You want to merge the cleanest of matches. In this scenario, limited keys produce only a few of key permutations, and a narrow search returns only extremely similar candidates. You can avoid overmatching the records.

The following table lists the recommended key generation and search levels for the scenarios:

Scenario	Key Generation Level	Candidate Search Level	Processing Time
Flight reservation	Extended	Extreme	More
Retail store	Standard	Typical	Optimal
Financial institution	Limited	Narrow	Less

Match population

Population is a definition of certain characteristics of data in the records that you want to match. Each supported country, language, or population has a match population.

When you ingress or import records, you can specify multiple populations for records. If the records don't contain population information, the population that's configured for the match model is used. For example, in your data with 25 records, 15 records are of the Japan population, and 10 records don't have any specified population. If the default population of the match model is Usa, the 10 records that do not have any specified population are treated as the Usa population during the matching process.

Before you configure populations when you ingress or import records, consider the following guidelines:

- Determine the type of match population to use.
- Select one or more populations that suit the data that you want to use for matching.
- Check whether the population supports the data that you want to use for matching.
- If you have mixed data from different languages, consider using multiple populations.

Merge strategy

A merge strategy indicates the action to be performed on the records that have gone through the match process. The process uses the merge thresholds, which are match score ranges, to determine the merge strategy. Based on the

merge strategy and the associated merge threshold of the rule, the record pairs result in automated, manual, threshold-based, or skipped merge.

The following table describes the merge strategies:

Merge Strategy	Description	Short Summary
Automated	Automatically merges the records identified as duplicates if their match scores fall within the specified merge threshold range for automated merge. For example, use the automated merge strategy for declarative rules that are based on unique identifier fields, such as social security number. Two people cannot have the same Social Security number, so you do not have to manually review the records.	Records are merged without review.
Manual	Queues the records identified as duplicates for manual merge if their match scores fall within the specified merge threshold range for manual merge. Users with the required privileges can review and merge the records in the Business 360 applications. For example, use the manual merge strategy for declarative rules that are based on fields, such as the address field. Multiple family members can share the same address and the records cannot be definite matches. You need to manually review the records to confirm whether these are duplicate records.	Humans review and then approve the merge.
Threshold-based	Available only when you select the fuzzy match strategy. Based on the match score of the record pairs, the match and merge process determines whether the result is automated, manual, or skipped merge. For example, use the threshold-based merge strategy for a declarative rule that is based on a field, such as address, that is quite similar but other field values are not. In this scenario, you can set the thresholds to skip merging records when the match score is 70 and lower, automate merging records when the match score is 90 and higher, and send for manual merge when the match score is from 71 to 89.	Match pairs are ignored.
Skip	The records in the pairs are not matched against each other if their match scores fall within the range set for the merge threshold for skipping merge.	Merge strategy is based on the specified thresholds for different strategies.

Merge threshold

Establish merge thresholds to control the level of similarity required for records to be considered a match. Adjust these thresholds based on the desired balance between precision and recall. A lower threshold results in higher recall and lower precision. A higher threshold results in higher precision and lower recall.

Guidelines to configure declarative rules

When you configure match rules, consider the following guidelines:

- Determine whether you need automated or a manual merge.

- Start with simple rules. Begin by creating simpler match rules before moving on to more complex ones. This approach allows you to refine and improve matching strategy based on the results.
- Create multiple match rules to cater to different scenarios and levels of matching accuracy. For example, you can have one match rule for exact matches, another for fuzzy matches, and additional rules for specific attributes or patterns. You can combine these rules using logical operators (AND, OR) to define the overall matching strategy.
- Don't have any unconditional fuzzy match rules. Always have some exact match filters on every match rule.
- Add one or more exact match fields to act as filters for each rule. Exact match fields can significantly improve performance by reducing the number of fuzzy match comparisons.
- Configure a match model with the least possible number of match rules and optimal match results.
- Create several almost identical rules with different exact filters to try to reduce the number of fuzzy matching. The filters can be attributes, such as postal code, phone number, or some other reliable values.

Segment matching

Use segment matching to limit matching to specific subsets of data based on the exact match field values. For example, you can create a subset of data based on country values. If you specify the segment value as Japan, the rule uses the records that have Japan as a country value for matching.

Test and validate match rules

Perform extensive testing and validation of your match rules using representative sample data sets. Evaluate the results to ensure that the rules are accurate, efficient, and meet your desired outcomes.

Tuning exercise

To ensure that the match rules work accurately, perform the following checks:

- Monitor and tune. Continuously monitor the performance of your match rules and periodically re-evaluate them to adapt to changing data patterns, business rules, and evolving requirements. Regular tuning can help maintain the accuracy and effectiveness of your match rules over time.
- Collaborate with data stewards. Involve data stewards and subject matter experts in the match rule creation and refinement process. Their domain knowledge and expertise can contribute valuable insights and improve the overall quality of your match rules.
- Leverage external data sources. Consider incorporating external data sources or reference data to enhance your match rules. External data, such as industry-specific data sets or third-party data providers, can provide additional context and improve matching accuracy.
- Pay attention to the suggestions provided by CLAIRE engine while configuring match rules.

Survivorship rules

In addition to match configuration, consider implementing survivorship configuration to determine how you can resolve conflicting data from matched records. Survivorship configuration defines which values to retain for specific attributes when multiple records merge.

After identifying match pairs, the merge process runs to consolidate the match pairs into master records. Survivorship is a process of creating a master records from multiple matching records. The goal is to create the best version of record by using the most reliable data available. Each field can have a different condition for deciding what is best.

Rules always override the source rank configuration. Decay only applies at the time of a merge, unmerge, or update event. Use logical grouping, when there are large number of source systems. Informatica Customer 360 or Default source system always provides highest trust with quickest decay for data steward changes.

Challenges

When you configure match and merge, you might face the following challenges:

- Similar data set. The match process identifies up to 10,000 candidates for each record.
- Transactional data or bad data. Separate the bad records by using Cloud Data Integration mappings.

Do's and don'ts of match and merge configuration

Consider the following do's and don'ts when you configure match and merge:

- When you ingress data, ensure that you avoid loading transactional data. Use Cloud Data Integration to filter out transactional data. Highly duplicate transactional data impacts performance negatively by generating excessive number of match comparisons.
- When you use threshold-based merge strategy, understand the rule outcomes for each scenario.
- When you use name and address for matching without unique identifiers, ensure that you configure tight thresholds to avoid false positives. For example, multiple family members can share the same address and the records can't be definite matches.
- When you use exact match rules, configure the appropriate fields for the candidate selection criteria.
- If required, regenerate the match keys after updating the candidate selection criteria. For example, if you change the default population, the match criteria values in a declarative rule change. Ensure that you regenerate the match keys for the records.
- Use the NOT_READY_FOR_MATCH flag to mark records to exclude from the match process. If a source record doesn't have enough information to contribute to the master record, you might want to mark it as not ready for match.
- During match tuning, perform match process first. You can rematch the records until you reach an optimal match configuration.
- During initial match tuning, use a smaller data set to test. Processing a smaller data set minimizes loss of time in the event of match job failure and allows for multiple tuning iterations. Ensure that you configure match rules to accommodate expected data volumes.
- Avoid choosing multiple candidate selection criteria unless it's essential, and you have such a requirement for your use case.
- Avoid manual match rules that use loose match level. You must not flood the data stewards with too many matches.

Training a match model

Before you train an ML model, you need to identify the training data set that is a true representation of your organization's data. The results produced by the ML model are only as good as the training data set the model is trained on.

Recommendations for selecting training data sets

The accuracy of a match model depends on the quality of the training data set. The training data set provides the examples that the machine learning (ML) model learns from. It is important that the training data set accurately reflects your business scenarios.

To select training data sets for training the ML model, review the following recommendations:

- Start by analyzing the characteristics of your data and the specific match requirements. Understand the data from where you want to select a training data set. Based on the understanding of your data, select a suitable training data set.
- For optimal training results, the data set sample must be similar to the production data. You can manually create a subset of production data by using the field values. For example, use the Country, State, or City code, or a combination of these to create a subset of production data.
- Ensure that the training data set contains the attributes that satisfy your matching requirements. For a relevant data set training, the data set sample must contain attributes same as the production data. For example, if the Phone Number and Email ID attributes are available in the production data, the same attributes must be available in the sample data set that is used for training.
- Ensure that the data sets contain some records that match and some records that do not match so that you have a clear idea of true positives and true negatives when you view the training results.
- For optimal training results, you can avoid duplicating data sets. If a data set has the same name, address, or any other key attributes in more than 25 records, it results in many duplicate data sets. Exclude some of the data to reduce duplicates.
- During the training process, the ML model learns the patterns and characteristics of the data set. Therefore, the training data set must be large enough for the ML model to understand the data patterns for making match predictions. For example, if you have a complex matching scenario, the ML model might need to use a larger number of match fields to effectively make match predictions. The need for a large number of match fields means that the model requires a larger data set.
- A data set must contain a minimum of 10,000 records. The optimal size data set can contain 50,000 to 100,000 records. A data set must contain less than 200,000 records.

Recommendations for training machine learning models

To train a machine learning (ML) model, review the following recommendations:

- You can train an ML model with a subset of production data. Consider training a match model in a non-production environment that does not contain the entire production data.
- A match model that you want to use for training must have the same population as the production data.
- During the training process, the training page displays the precision, recall, and accuracy values that indicate the learning of the ML model. Consider reviewing the table in between the batches during the training process.
- When the precision, recall, and accuracy values display a steady increase for an ML model during the training of several batches, it indicates that the ML model is completely trained and you can stop training the ML model.
- For an optimal ML model training, you might want to train the model for at least 20 batches.

Match and merge best practices checklist

The following checklist summarizes the best practices for configuring match and merge:

- Know your data.
- Define candidate selection criteria.

- Set the maximum candidate limit.
- Define fuzzy match fields.
- Define exact match fields.
- Set the merge threshold limits.
- Leverage segment matching.

Author

Informatica Global Customer Support